

EduVerse LMS – Project Overview

Introduction

EduVerse LMS is a next-generation **School Management System (ERP) + Learning Management System (LMS) + Gamified Computer Education Platform** designed specifically for schools.

The platform aims to solve two major problems:

1. Schools need a complete digital management system for administration.
2. Students need an engaging and interactive way to learn computer education.

EduVerse combines both into a single cloud-based platform where schools can manage operations while students enjoy a game-like learning experience.

Project Vision

To create India's most engaging school learning platform that transforms traditional computer education into a fun, interactive, and rewarding experience while providing schools with a complete management solution.

Project Objectives

For Schools

Provide a centralized platform to manage:

- Student Records
- Teacher Records
- Attendance
- Fees
- Library
- Bus Transportation
- Reports & Analytics
- School Communication

For Students

Provide a gamified learning environment through:

- Video Lessons
- Interactive Quizzes
- Mini Games
- XP & Level System
- Achievements & Badges
- Leaderboards
- Progress Tracking

For Parents

Provide transparency through:

- Attendance Monitoring
- Academic Progress
- Fee Status
- Notifications

Platform Structure

The platform is divided into two major systems:

1. School Management System (ERP)

Used by:

- School Admin
- Teachers
- Parents

Modules include:

- Student Management
- Attendance Management
- Fee Management
- Library Management
- Bus Management
- Teacher Management
- Reports & Analytics

2. Computer Learning Platform (LMS)

Managed by:

- Super Admin

Accessed by:

- Students
- School Admins

Features include:

- Grade-wise Courses
 - Video Lessons
 - Assessments
 - Certificates
 - Progress Tracking
 - Gamification
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User Roles

Super Admin

Platform Owner

Responsibilities:

- Manage Schools
 - Manage Courses
 - Set School Pricing
 - Upload Content
 - View Platform Analytics
 - Generate Reports
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School Admin

Responsibilities:

- Manage Students
 - Manage Teachers
 - Manage Attendance
 - Manage Fees
 - Manage Library
 - Manage Teacher Salary
 - View Student Course Progress
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Teacher

Responsibilities:

- Mark Attendance
 - View Students
 - Assign Work
 - Track Progress
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Student

Responsibilities:

- Learn Courses
 - Complete Assessments
 - Earn Rewards
 - Track Progress
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Parent

Responsibilities:

- Monitor Child Progress
- Track Attendance
- Check Fee Status

Gamification Features

XP System

Students earn points by:

- Watching Videos
- Completing Quizzes
- Daily Logins
- Achievements

Levels

Students progress through levels:

- ☐ Seedling
- ☐ Explorer
- ☐ ☐ Adventurer
- ☐ Champion

Badges

Examples:

- First Lesson
 - Quiz Master
 - Perfect Score
 - Fast Learner
 - Weekly Champion
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Coins & Rewards

Students earn virtual coins that can be used to:

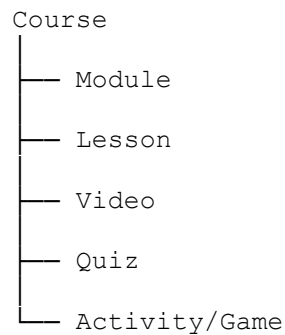
- Unlock Avatars
- Customize Profiles
- Unlock Themes

Leaderboards

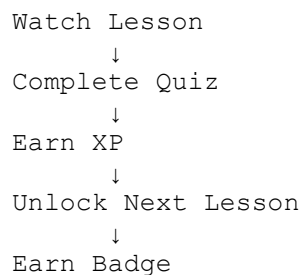
Students compete within their class.

This creates healthy competition and motivation.

Course Structure



Learning Flow



Students cannot skip lessons.

Learning follows a structured path.

□ Technology Stack

Frontend

- Next.js 14
- TypeScript
- Tailwind CSS
- Shadcn UI
- Framer Motion

Backend

- Node.js
- Express.js
- Prisma ORM

Database

- PostgreSQL

Authentication

- JWT
- NextAuth

Storage

- AWS S3 / Cloudflare R2

Deployment

- Vercel
 - Railway / Render
 - Cloudflare CDN
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☐ **Security Features**

- Role-Based Access Control (RBAC)
 - Password Encryption (bcrypt)
 - JWT Authentication
 - School Data Isolation
 - Rate Limiting
 - Secure File Uploads
 - HTTPS Enforcement
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☐ **Platform Accessibility**

Desktop

- Super Admin
- School Admin
- Teachers

Mobile

- Students
- Parents

Tablet Optimized

Specially designed for students in Grades 1–5 with:

- Cartoon UI
 - Large Buttons
 - Interactive Elements
 - Kid-Friendly Navigation
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☐ **Expected Benefits**

Schools

- Reduced administrative workload
- Better student tracking
- Centralized operations

Students

- Increased engagement
- Better learning outcomes
- Consistent motivation

Parents

- Better visibility into child performance

Platform Owner

- Multi-school SaaS revenue model
 - Scalable architecture
 - Centralized course management
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Conclusion

EduVerse LMS is not just a School ERP or LMS. It is a complete digital ecosystem that combines school management, computer education, and gamified learning into a single scalable platform. The goal is to make learning enjoyable, measurable, and impactful while providing schools with powerful administrative tools.